



STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

Lab Teacher: Dr. Aymen TRIGUI

Course Code: 462 CCS-3

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Section Number: 904/907

S. No

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LAB ACTIVITY - 9

(1.5 Marks)

Objective: Apply the K-means to cluster the generated dataset.

The following tasks can be performed:

- Generate the dataset.
- Visualize the clusters (using the function PLSDData provided in the code, it is based on plt.scatter)
- Applying the K-means to cluster data.
- Measure the clustering quality by the Silhouette score.

In this step number of clusters (K) can be modified as an hypermeter

Clustering on Normalized data.

- Normalize the data and cluster by K-means.
- Compare the obtained clustering with clustering on data without normalization.
- Comment on the necessity of normalization

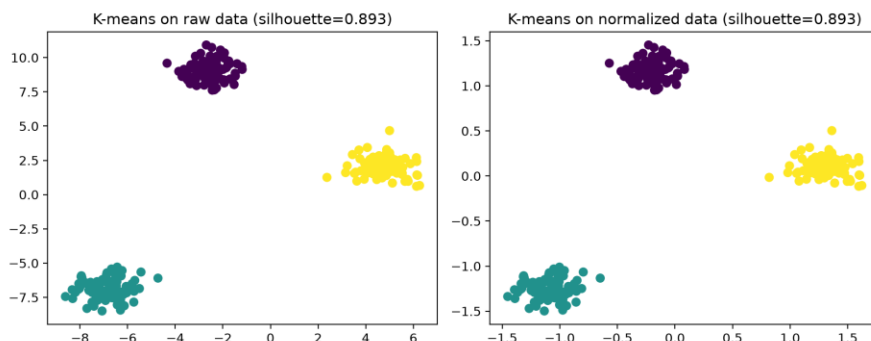
In fact, some features are very important such as the mass feature for the fruit. After normalization the mass loses its importance, and the clustering is worse.

Generated dataset with 3 clusters.

Silhouette score on raw data: 0.8932

Silhouette score on normalized data: 0.8934

Comment: Normalization helps when features have different scales because it gives each feature equal influence during clustering.



Marks Obtained	/1.5
Teacher's Signature:	